Luke Lopez

915-258-6597 — lukelopez499@gmail.com — College Station, TX — US Citizen

Selected ASIC Projects

Custom CMOS Standard Cell + Post-Layout Extraction (Cadence Virtuoso)
Designed and laid out CMOS logic cell achieving DRC/LVS clean verification. Performed parasitic RC extraction and analyzed schematic vs post-layout delay and power tradeoffs.

Custom 8-Bit ALU IC Design (Verilog RTL)
Designed arithmetic unit using Carry-Lookahead and Ripple-Carry architectures; evaluated speed vs area tradeoffs. Implemented datapath and control logic with structured verification.

Single-Cycle ARM Processor (Verilog RTL)
Developed custom CPU including ALU, register file, control unit, and memory interface. Explored architectural timing tradeoffs and validated instruction execution via simulation.

5-Stage Pipelined ARM Processor (Verilog RTL)
Implemented hazard detection, forwarding logic, and stall control. Optimized datapath timing and validated pipeline behavior through structured simulation.

Fully Differential Op-Amp (Cadence Virtuoso)
Designed transistor-level amplifier achieving 2 GHz unity-gain bandwidth with 50° phase margin. Performed small-signal analysis, compensation tuning, and stability validation.

FPGA Hardware Accelerator + Linux Driver Integration
Implemented register-mapped accelerator and integrated with Linux using custom device driver. Performed cross-layer debugging between RTL and OS.

Technical Skills

HDL / RTL: Verilog, VHDL, combinational & sequential logic, FSM design, datapath architecture, pipelining, testbench development
ASIC & VLSI: Static CMOS, pass-transistor logic, propagation delay analysis, parasitic extraction, timing constraints, power–performance–area tradeoffs
Tools: Cadence (Virtuoso), LTSpice, ModelSim, FPGA toolchains, Altium Designer
Programming: C/C++, Python, ARM Assembly
Hardware: FPGA development, board-level debugging, embedded systems integration

Work Study Experience

IT Support Technician – Texas A&M University
- Provided hardware and network troubleshooting across 100+ classroom systems.
- Managed AV control systems for class infrastructure.

AI-Solar Capstone – Team Lead
- Led cross-disciplinary team developing AI-based Maximum Power Point Tracking achieving 99.4% IV-curve accuracy.
- Designed predictive models for panel orientation achieving 97.2% optimization accuracy.
- Coordinated integration between embedded hardware, firmware, and software components.

Research Assistant – AI in Classroom
- Analyzed AI usage in laboratory's and redesigned workflows.
- Increased lab completion rates by 20%.

Embedded Systems & Hardware Integration
- Hardware Debug & Validation: oscilloscope and logic analyzer-based debugging of timing faults, signal integrity issues, and communication mismatches.
- Communication Protocols: implemented and debugged UART, SPI, and I2C interfaces across embedded subsystems.

Relevant Coursework

ECEN 454 – Digital Integrated Circuit Design
Timing analysis, setup/hold constraints, combinational and sequential logic synthesis.

ECEN 326 – Electronic Design Laboratory
Two-stage amplifier design, Miller compensation, stability analysis.

ECEN 248 – Integrated Circuit Design
CMOS layout, noise margins, delay modeling, static/dynamic logic fundamentals.

ECEN 350 – Computer Architecture & Design
Pipelining, hazard resolution, memory hierarchy, architectural performance analysis.

ECEN 325 – Electronics
Small-signal modeling, transistor-level amplifier design, frequency response.

ECEN 314 – Signals & Systems
Linear systems, Laplace transforms, frequency-domain behavior, stability criteria.

Education

Bachelor of Science in Computer Engineering May 2026
Texas A&M University